
KTU
NOTES

d) dual converter.

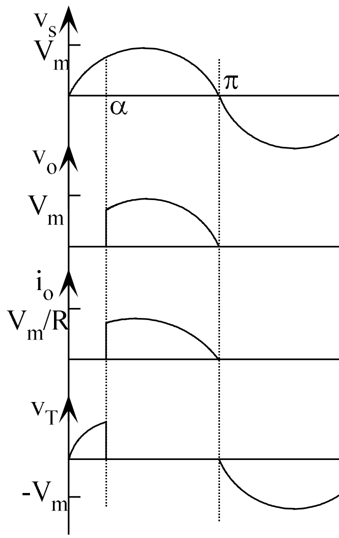
A semiconverter is one quadrant converter and it has one polarity
current.

A full converter is a two quadrant converter and the polarity of the
either positive or negative; but the output current has one polarity only
unidirectional switch).

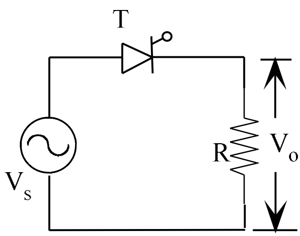
A dual converter can be operated in 4 quadrants; and both the output
be either positive or negative.

SINGLE PHASE HALF-WAVE CONTROLLED RECTIFIER WITH R I

Assumptions :- 1) Source voltage v_s is pure sinusoidal.
2) Devices are ideal (no voltage drop during ON state)
3) Source inductance is negligible



In the positive half cycle, thyristor is in forward blocking state and it is turned on at an angle $\omega t=\alpha$ by applying a gate pulse. For $\alpha<\omega t<\pi$, load voltage is equal to the source voltage ie. $v_o=v_s$. Load current v_o/R is in phase with load voltage. At $\omega t=\pi$, load current (hence thyristor current) falls to zero and it is turned off (line or natural commutation - when thyristor current falls below holding current it is turned off). No additional circuit is required for commutation.



Instantaneous value of source voltage, $v_s = V_m \sin \omega t$

Output dc power, $P_{dc} = V_{dc} \times I_{dc}$

Output ac power, $P_{ac} = V_{rms} \times I_{rms}$

Rectification efficiency, $\eta = \frac{P_{dc}}{P_{ac}} \times 100\%$

Transformer utilization factor, $TUF = \frac{P_{dc}}{V_s \times I_s}$

RMS value of source voltage, $V_s = \frac{V_m}{\sqrt{2}}$

RMS value of source current, $I_s = I_{rms}$

Peak Inverse Voltage (Reverse blocking voltage) $PIV = V_m$

Displacement angle, Φ = angle between the fundamental source voltage and fur
current = angle between V_{s1} and I_{s1}.
V_{s1} and I_{s1} = RMS value of the fundamental source voltage and fundamental so
respectively
Since source voltage is sinusoidal, $V_{s1}=V_s$.
Displacement factor or Displacement power factor DPF= cosΦ

Distortion factor, $DF = \frac{I_{s1}}{I_s}$ (measure of distortion of the waveform) (for sine

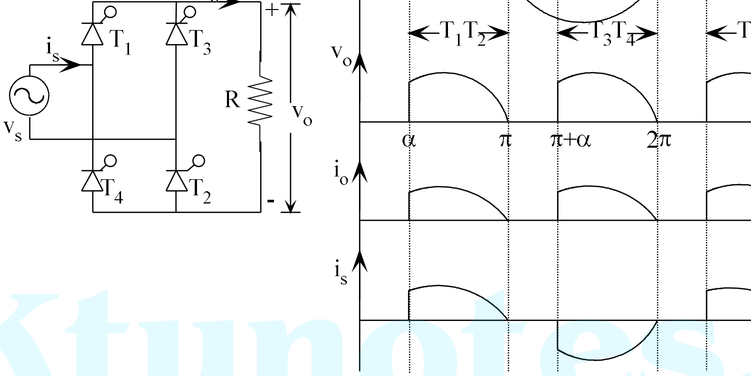
Harmonic factor HF or Total Harmonic Distortion THD, $THD = \frac{\sqrt{I_s^{\,2} - I_{s1}^{\,2}}}{I_{s1}}$ (F

Input power factor = $PF = \frac{V_{s1} I_{s1} \cos\phi}{V_s I_s} = \frac{I_{s1}}{I_s} \times \cos\phi = DF \times DPF$

Distortion factor, $DF = \frac{1}{\sqrt{1 + THD^2}}$

Example 1: If a 1-ph half-wave converter has a purely resistive load of R
$\alpha = \pi / 2$, determine a) the rectification efficiency b) the form factor FF c) t
TUF and e) PIV of the thyristor.

Solution : V_{dc}=0.1592V_m I_{dc}=0.1592V_m/R V_{rms}=0.3536V_m I_{rms}=0.3536V_m/
$P_{dc}=(0.1592V_m)^2/R$ $P_{ac}=(0.3536)^2/R$
FF=2.221 RF=1.983 V_s=0.707V_m TUF=0.1014 PIV=V_m



When T_1 and T_2 are conducting, source current is same as the load current a
conducting source current is negative of the load current.

SINGLE PHASE FULL CONVERTER WITH RL-LOAD (Discontinuous

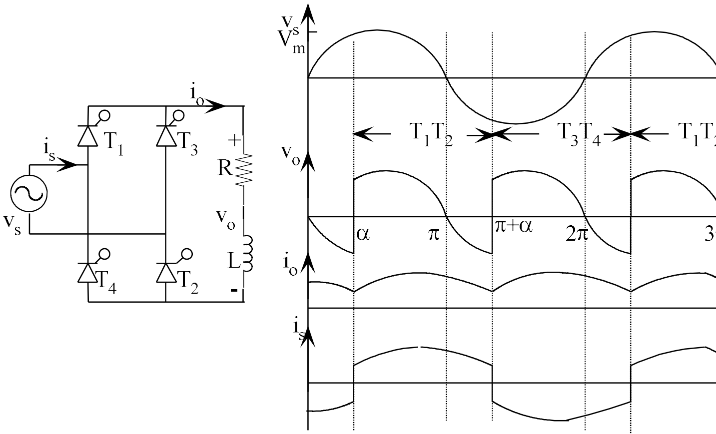
During positive half cycle, thyristors T_1 and T_2 are in forward blocking mode are turned on at $\omega t=\alpha$ by applying gate pulses simultaneously. The load is supply through T_1 and T_2. At $\omega t=\pi$, T_1 and T_2 will continue to conduct due to inductance. A part of the stored energy is fed back to the source and second load resistance. At $\omega t=\beta$, stored energy decreases to zero and T_1 and T_2 are turn

During negative half cycle, T_3 and T_4 are in forward blocking state and simultaneously at $\omega t=\pi+\alpha$.

Average output voltage, $V_{dc} = \frac{1}{\pi}\int_{\alpha}^{\beta} V_m \sin\omega t d(\omega t) = \frac{V_m}{\pi}(\cos\alpha - \cos\beta)$

FULL CONVERTER WITH RL-LOAD (Continuous conduction)
(Load current is continuous)

During positive half cycle, thyristors T_1 and T_2 are in forward blocking mode are turned on at ωt=α by applying gate pulses simultaneously. The load is supply through T_1 and T_2. At ωt=π , T_1 and T_2 will continue to conduct due to inductance. A part of the stored energy is fed back to the source and second load resistance. At ωt=π+∝ , T_3 and T_4 are turned on and T_1 and T_2 automat reverse voltage applied across its anode and cathode. Here, the inductance is s the load current is continuous.



Average output voltage, $V_{dc} = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin\omega t d(\omega t) = \frac{2V_m}{\pi}\cos\alpha$ (Note: This equ continuous conduction mode)

For sufficiently large value of inductance, the load current may be assumed to b ripple free (i_o=I_o where i_o is the instantaneous value and I_o is the average value

Note : For an RL-load, average voltage is always positive. Average voltage is z

Note : If α is greater than 90°, continuous conduction is not possible for an RL

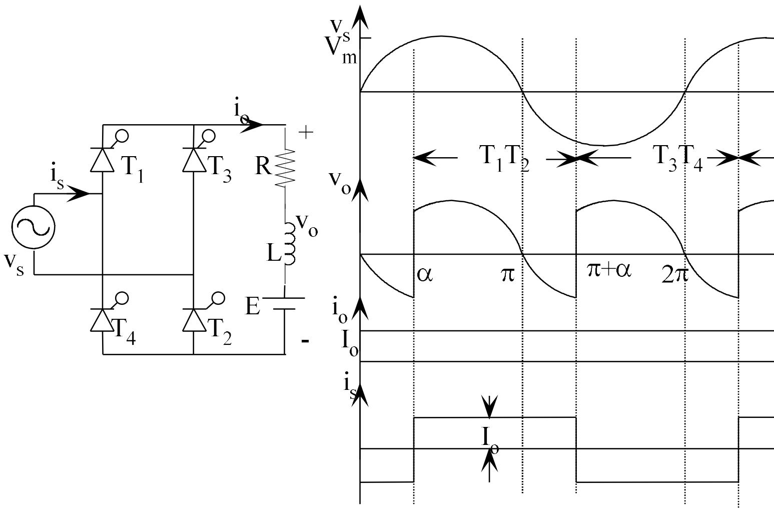
$\alpha \le \Phi$, conduction is continuous where Φ is the impedance angle ($\Phi = \tan^{-1}(\frac{\omega L}{R}$

becomes discontinuous.

FULL CONVERTER WITH RLE-LOAD (Continuous conduction)

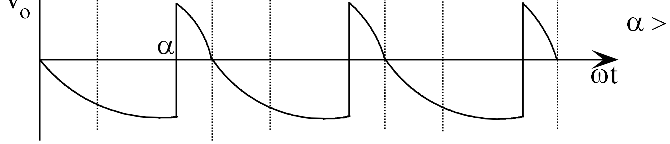
Controlled rectifiers are commonly used for the speed control of dc moto voltage is controlled to vary the speed. DC motor is an example of RLE load armature resistance and L represents the armature winding inductance and E re

Assumption : Load current is continuous and ripple free $i_o = I_o$



Average output voltage, $V_{dc} = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin\omega t d(\omega t) = \frac{2V_m}{\pi}\cos\alpha$

If $\alpha < 90^o$, V_{dc} is positive (rectification); $\alpha = 90^o$, V_{dc} is zero and if $\alpha > 90^o$, V_{dc}

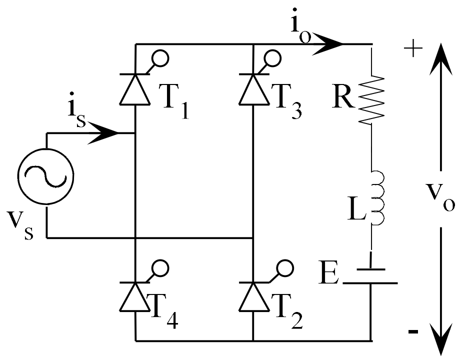


Depending on the value of α, the average output voltage can be either positive o
provides two quadrant operation.

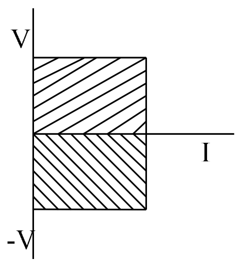
For α > 90°, average output voltage is negative, but the load current continu
direction, so output power is negative. Power flows from load to source. It is p
load, i.e., when the load has an emf source (RLE). Operation of the conv
quadrants.

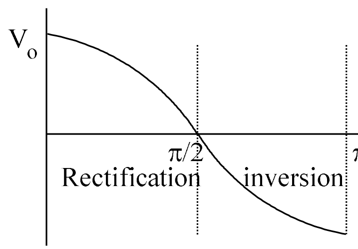
(For regenerative braking of dc motor, α is made greater than 90° and at th
terminals are interchanged so that the current flows from the dc machine, actin
ac source. A mechanical arrangement is used for reversal of the connections.)

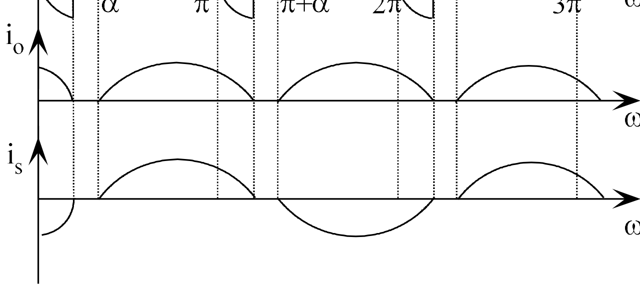
Connection of converter and DC motor during inversion (α > 90°) is shown bel



Note: Inversion is not possible for passive load (RL load) even if α > 90°.






α
π
π+α
2π
3π
ω
i_o
ω
i_s
ω

$a_o = \frac{2}{T}\int_0^{T/2} f(t)dt$ $a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega t dt$ $b_n = 0$	$a_o =$ $a_n =$ $b_n = \frac{4}{T}\int_0^{T/2} f(t)$
For half-wave symmetry, $f(t) = -f(t+T/2)$ $a_o = 0$ $a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega t dt$; n odd $b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega t dt$; n odd	For quarter-wav $f(t) = f(-t)$ & $f(t) = -$ $a_o = 0$; b $a_n = \frac{8}{T}\int_0^{T/4} f(t)\cos$ $f(t) = -f(-t)$ & $f(t) = -$ $a_o = 0$; a $b_n = \frac{8}{T}\int_0^{T/4} f(t)\sin$

$$\cos(A-B) - \cos(A+B) = 2\sin A \sin B$$
$$\sin 2A = 2\sin A\cos A$$
$$1 + \cos 2A = 2\cos^2 A$$

Analysis of 1-phase full converter with RLE load

Distortion Factor DF, Displacement Factor DPF and Input Power Factor I

Assumption :

i) Load current is continuous and ripple free $i_o = I_o$

ii) Source voltage is assumed to be sinusoidal

V_s = RMS value of source voltage

V_{s1} = RMS value of fundamental source voltage

Since the source voltage is sinusoidal, $V_{s1} = V_s$.

I_s = RMS value of source current

I_{s1} = RMS value of fundamental source current

Source current, $i_s(t) = a_o + \sum_{n=1}^{\infty}[a_n \cos n\omega t + b_n \sin n\omega t]$

$i_s(t)$ has half-wave symmetry. Hence, a_o=0.

$$a_1 = \frac{4}{2\pi}\int_{\alpha}^{\pi+\alpha} I_o \cos \omega t d\omega t = \frac{-4I_o}{\pi}\sin\alpha$$

$$b_n = \frac{4}{2\pi}\int_{\alpha}^{\pi+\alpha} I_o \sin \omega t d\omega t = \frac{4I_o}{\pi}\cos\alpha$$

$$\phi_1 = \tan^{-1}\frac{a_1}{b_1} = \tan^{-1}(\frac{-\sin\alpha}{\cos\alpha}) = -\alpha$$

Maximum value of fundamental source current = $\sqrt{a_1^{\,2} + b_1^{\,2}}$

RMS value of fundamental source current, $I_{s1} = \frac{\sqrt{a_1^{\,2} + b_1^{\,2}}}{\sqrt{2}} = \frac{2\sqrt{2}I_o}{\pi} = 0.9I_o$

RMS value of source current, $I_s = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} I_o^{\,2} d\omega t} = I_o$

Displacement Factor, DPF= $\cos\Phi_1$ = $\cos(-\alpha)$

Distortion factor, $DF = \frac{I_{s1}}{I_s} = \frac{2\sqrt{2}}{\pi} = 0.9$

Input power factor = $PF = \frac{V_s I_{s1}\cos\phi}{V_s I_s} = \frac{I_{s1}}{I_s} \times \cos\phi = \frac{2\sqrt{2}}{\pi}\cos\alpha$

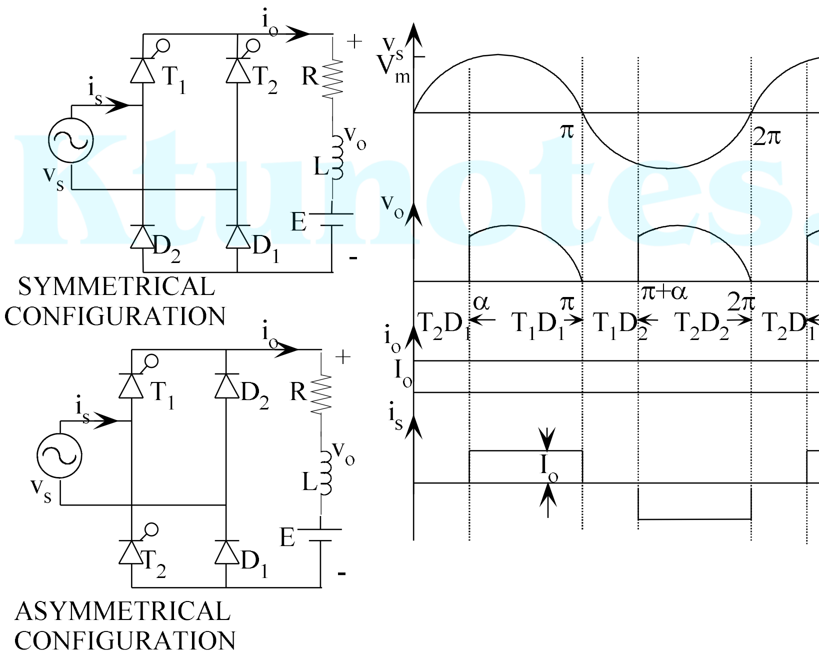
Harmonic factor HF or Total Harmonic Distortion THD, $THD = \frac{\sqrt{I_s^{\,2} - I_{s1}^{\,2}}}{I_{s1}} = ($

Note :

$\phi_1 = -\alpha$	$I_{s1} = \frac{2\sqrt{2}I_o}{\pi} = 0.9$		
DPF= $\cos\Phi$ lag	$DF = \frac{I_{s1}}{I_s}$	$PF = \frac{I_{s1}}{I_s} \times \cos\phi$	T

Average output voltage , $V_{dc} = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin\omega t d(\omega t) = \frac{V_m}{\pi}(1+\cos\alpha)$

Average output voltage is never negative; it cannot operate as inverter.



Source current, $i_s(t) = a_o + \sum_{n=1}^{\infty}[a_n \cos n\omega t + b_n \sin n\omega t]$

$i_s(t)$ has half-wave symmetry. Hence, a_o=0.

$$a_1 = \frac{4}{2\pi}\int_{\alpha}^{\pi} I_o \cos\omega t d\omega t = \frac{-2I_o}{\pi}\sin\alpha$$

$$b_1 = \frac{4}{2\pi}\int_{\alpha}^{\pi} I_o \sin\omega t d\omega t = \frac{2I_o}{\pi}(1+\cos\alpha)$$

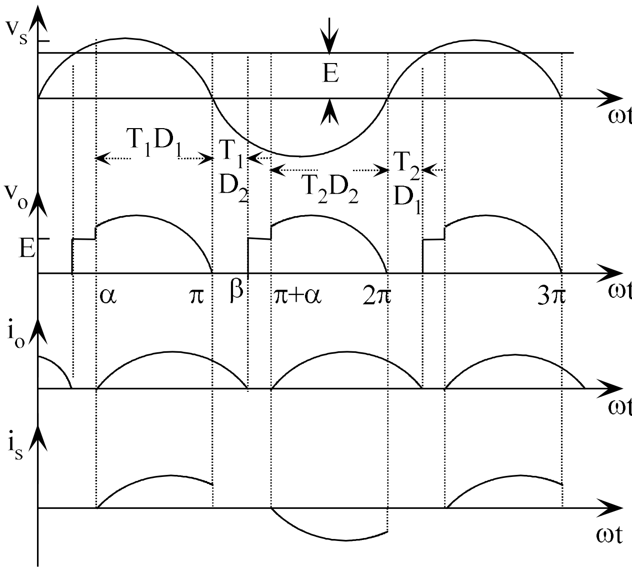
Maximum value of fundamental source current,

$$\sqrt{a_1^2 + b_1^2} = \frac{2I_o}{\pi}\sqrt{\sin^2\alpha + (1+\cos\alpha)^2} = \frac{2\sqrt{2}I_o}{\pi}\sqrt{1+\cos\alpha} = \frac{4I_o}{\pi}\sqrt{\cos^2\frac{\alpha}{2}} = \frac{4I_o}{\pi}\cos$$

2	π 2		
DPF= cosΦ	$DF = \dfrac{I_{s1}}{I_s}$	$PF = \dfrac{I_{s1}}{I_s} \times \cos\phi$	

Discontinuous conduction

T_1 will be in forward blocking state and D_1 will be forward biased only a $V_m \sin\omega t$ is equal to battery emf E. When all the devices are off (or load cu voltage will be equal to E. T_1 and D_1 start conducting from α and the output v the source voltage. But, at ωt=π, D_2 is forward biased and D_1 is reverse biase freewheels through T_1 and D_2 till ωt=β. Load voltage will be zero from ωt=π to load voltage is not zero. From ωt=β to ωt=π+α, not devices are conducting an as the battery emf E. When T_1 and D_1 are conducting, source current is sa when T_1 and D_2 are conducting, source current is zero.



Example 1 : *A 1-phase full wave semiconverter is used to charge a 24V batter connected in series with the battery to limit the current. A large inductor is Find the output power and input p.f. for α = 45°. Input voltage is 50V. Assume free current.*

Solution : $V_{dc} = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin\omega t d(\omega t) = \frac{V_m}{\pi}(1+\cos\alpha)$ = 207V

$R_L = \frac{V_{dc}^{\ 2}}{P_o}$ =42.85Ω

For 0.5kW, required voltage, $V_{dc} = \sqrt{P_o \times R_L}$ = 146.4V=$\frac{V_m}{\pi}(1+\cos\alpha)$

$\alpha = 65.54°$

Example 3 : *Power supplied to a 230V, 1kW resistive load is to be controlled converter and b) semiconverter. Find the following quantities for both types output. i) Average voltage ii) RMS value of input current iii) fundamental com iv) displacement factor v) distortion factor vi) THD and vii) input power factor Assume continuous and ripple free load current.*

$R_L = \frac{V_{dc}^{\ 2}}{P_o}$ =52.9Ω

a) FULL CONVERTER

For 500W output,

$V_{dc} = \sqrt{P_o \times R_L}$ =162.6V=$\frac{2V_m}{\pi}\cos\alpha$

$\alpha = 38.24°$

$I_o = \frac{V_{dc}}{R}$ =3.073A

$\phi_1 = -\alpha$ = -38.24°	$I_{s1} = \frac{2\sqrt{2}I_o}{\pi}$ =2.766A		
DPF= cosΦ=0.785	$DF = \frac{I_{s1}}{I_s}$ =0.9	$PF = \frac{I_{s1}}{I_s} \times \cos\phi$ =0.707	*THD*

b) SEMICONVERTER

For 500W output,

$V_{dc} = \sqrt{P_o \times R_L}$ =162.6V=$\frac{V_m}{\pi}(1+\cos\alpha)$

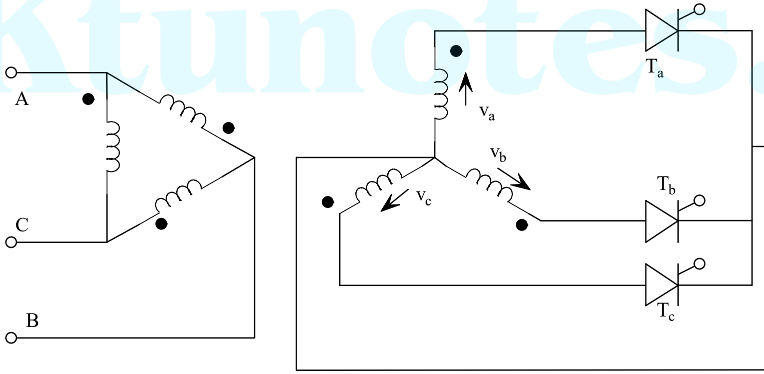
$\alpha = 55.2°$

The three thyristors are in common cathode configuration, hence the thyris
anode voltage will conduct if a firing pulse is applied.

T_a can be turned on only after ωt=30°. Hence, the reference point for firin
$\omega t = 30°$. The reference points for T_b and T_c are $\omega t = 150°$ and $\omega t = 270°$ respe

3-pulse converter with R load

If $\omega t \le 30°$, load current is continuous and if $\omega t > 30°$, current is discontinuou



Continuous conduction ($0 \le \alpha \le 30°$)

$$V_{dc} = \frac{3}{2\pi}\int_{\alpha+30}^{\alpha+150} V_m \sin\omega t d(\omega t)$$

$$= \frac{3}{2\pi}V_m\left\{-\cos\omega t\right\}_{\alpha+30}^{\alpha+150}$$

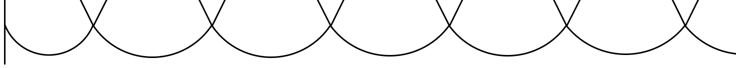
$$= \frac{3}{2\pi}V_m\left\{\cos(\alpha+30)-\cos(\alpha+150)\right\}$$

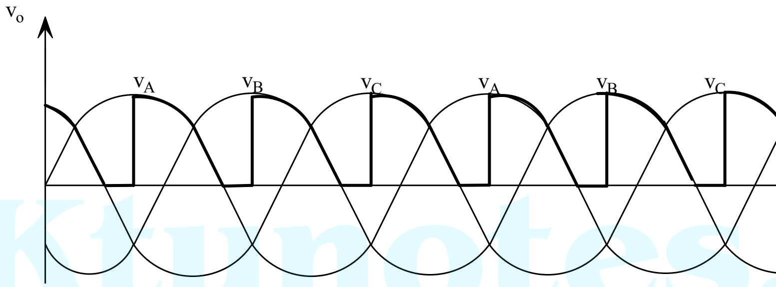
$$= \frac{3}{2\pi}V_m\left\{\cos(\alpha+90-60)-\cos(\alpha+90+60)\right\}$$

$$= \frac{3}{2\pi}V_m \times 2\sin(\alpha+90)\times\sin(60)$$

$$= \frac{3}{2\pi}V_m \times 2\cos\alpha\times\frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2\pi}V_m\cos\alpha$$

$$V_{dc} = \frac{3\sqrt{3}}{2\pi}V_m\cos\alpha$$





3-PHASE FULLY CONTROLLED BRIDGE RECTIFIER
(FULL CONVERTER OR SIX PULSE CONVERTER)

3-phase half-wave converter is rarely used because it introduces dc componen which leads to saturation of the magnetic core of the transformer. It has relative

In the six-pulse converter, each thyristor conducts for 120°. Input curr output has 6 pulses/cycle. All the six thyristors are divided into two groups. Up produces positive pulses in the output. Lower group (T_2, T_4 & T_6) produces neg

Upper group can be considered as a 3-pulse converter with common and lower group can be treated as a 3-pulse converter with common anode conf

A thyristor from upper group can be turned on when its anode voltage is two phase voltages. Firing angle of T_1 is measured from 30° (reference point) o

Reference angle for firing of T_1, T_3 & T_5 are 30°, 150° and 270° respective

A thyristor from lower group can be turned on when its cathode is most ne for firing T_2, T_4 & T_6 are 90°, 210° and 330° respectively.

When a thyristor of upper group is conducting, then one thyristor from lov for return path.

3-PHASE FULL CONVERTER WITH RLE LOAD (CONTINUOUS CON

Assumption Load current is assumed to be continuous and ripple free (i_o=I_o)

sequence. T_1 will conduct for 120° and it is turned off automatically when T_3 is

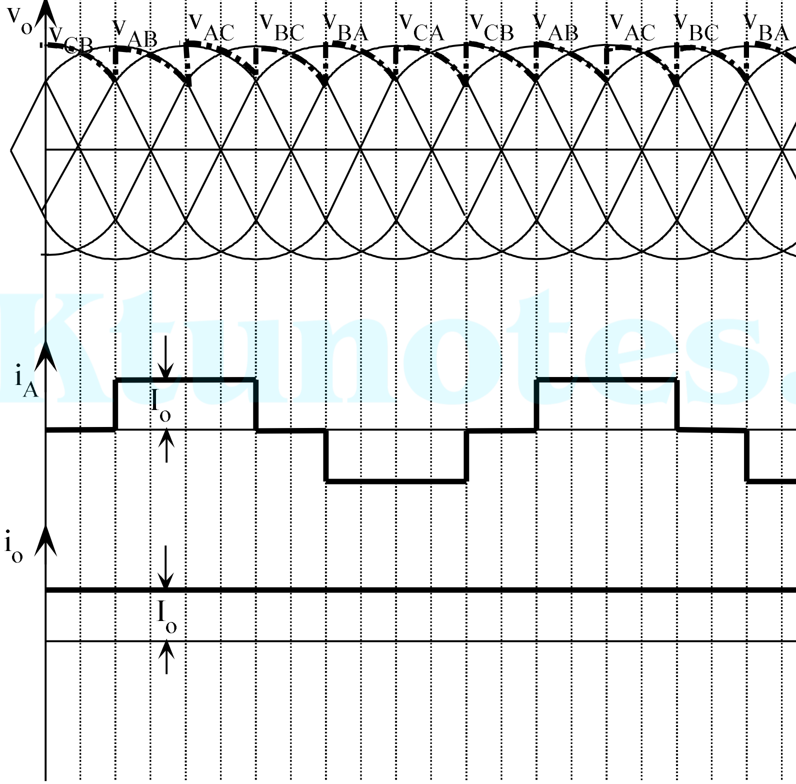
Device	Instant of turning ON	Instant of turning OFF
T_1	$\alpha + 30°$	$\alpha + 30° + 120°$
T_2	$\alpha + 90°$	$\alpha + 90° + 120°$
T_3	$\alpha + 150°$	$\alpha + 150° + 120°$
T_4	$\alpha + 210°$	$\alpha + 210° + 120°$
T_5	$\alpha + 270°$	$\alpha + 270° + 120°$
T_6	$\alpha + 330°$	$\alpha + 330° + 120°$

Let firing angle be α = 30°. Since reference angle for T_1 is 30°, T_1 is turned on
T_3, T_4, T_5 and T_6 are fired at $\omega t = 120°, 180°, 240°, 300°, 360°$. Since each d
120°, T_1 stops conducting at $\omega t = 60 + 120° = 180°$.

Device	Instant of turning ON	Instant of turning OFF
T_1	60°	180°
T_2	120°	240°
T_3	180°	300°
T_4	240°	360°
T_5	300°	420°
T_6	360°	480°

During 0-60°, T_5 and T_6 are conducting and the load voltage will be $v_o = v_C - v_B =$
voltages in each duration are as shown below.

Duration	Devices ON	Load voltage
0-60°	T_5T_6	v_{CB}
60°-120°	T_1T_6	v_{AB}
120°-180°	T_1T_2	v_{AC}
180°-240°	T_3T_2	v_{BC}
240°-300°	T_3T_4	v_{BA}
300°-360°	T_5T_4	v_{CA}



AVERAGE OUTPUT VOLTAGE - Continuous Conduction

Note : If $v_A = V_m \sin(\omega t)$ & $v_B = V_m \sin(\omega t - 120°)$, $v_{AB} = \sqrt{3} V_m \sin(\omega t + 30°)$

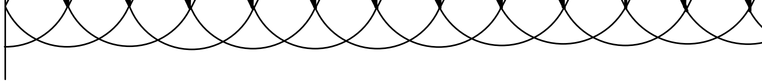
Average value of output voltage is

$$V_{dc} = \frac{6}{2\pi} \int_{\alpha+30}^{\alpha+90} \sqrt{3} V_m \sin(\omega t + 30°) d(\omega t)$$

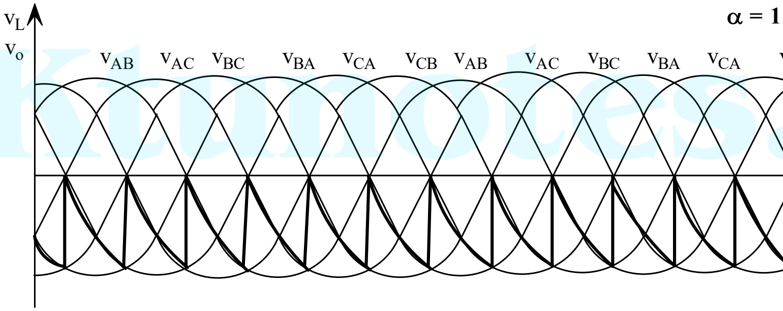
$$= \frac{6\sqrt{3}V_m}{2\pi} [-\cos(\omega t + 30°)]_{\alpha+30°}^{\alpha+90°}$$

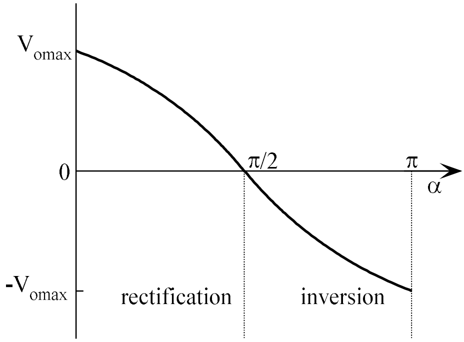
$$= \frac{6\sqrt{3}V_m}{2\pi} [\cos(\alpha + 60°) - \cos(\alpha + 120°)]$$

$$= \frac{6\sqrt{3}V_m}{2\pi} [\cos(\alpha + 90° - 30°) - \cos(\alpha + 90° + 30°)]$$



For RLE load, if α is greater than 90°, average value of output voltage is nega
is positive, power will flow from load to source. This is called inversion. Inve
the load is RLE type (active load) and α is greater than 90°.





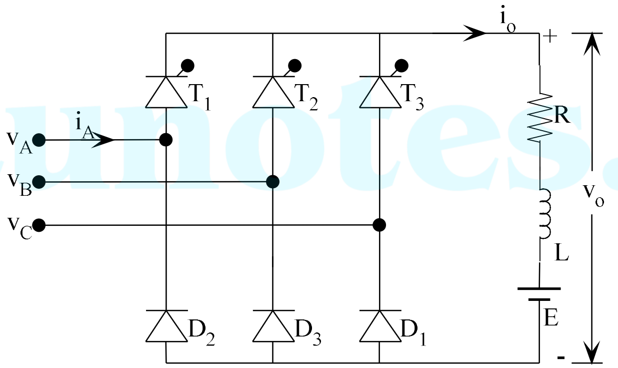
Line Commutated Inverter:-

3-PHASE HALF CONTROLLED BRIDGE RECTIFIER (3-PHASE SEM

Freewheeling mode of operation of bridge connected rectifiers can be realized l
thyristors with diodes.

3-phase semiconverter with RLE load

Assumption : Load current is continuous and ripple free

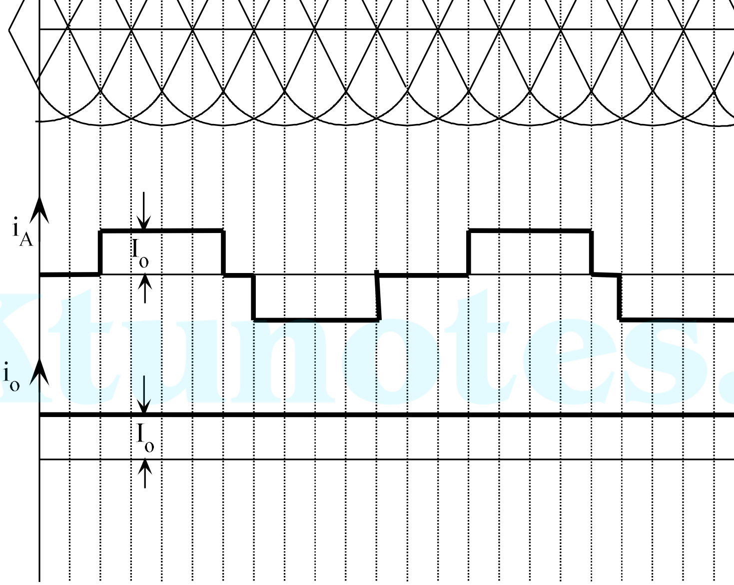


Note : A diode will be forward biased when its cathode is most negative; th
biased from 90° to 210°. D_2 will be forward biased from 210° to 330°. D_3 is fo
to 450°. (Each diode conducts for 120°)

Assume α = 30°

At α = 30° (ωt=60°) T_1 is turned on. At that instant D_3 is forward biased,
through T_1 and D_3, and load voltage is equal to v_{AB}. At ωt=90°, D_1 becomes fo
transferred from D_3 to D_1. As a result v_{AC} appears across the load. T_1 con
ωt=180°. At this point, T_2 is turned on, which commutates T_1. As a result, volta
30°.

α = 30°		
0-60°	T_3D_3	v_{CB}
60°-90°	T_1D_3	v_{AB}
90°-180°	T_1D_1	v_{AC}
180°-210°	T_2D_1	v_{BC}
210°-330°	T_2D_2	v_{BA}
330°-360°	T_3D_2	v_{CA}

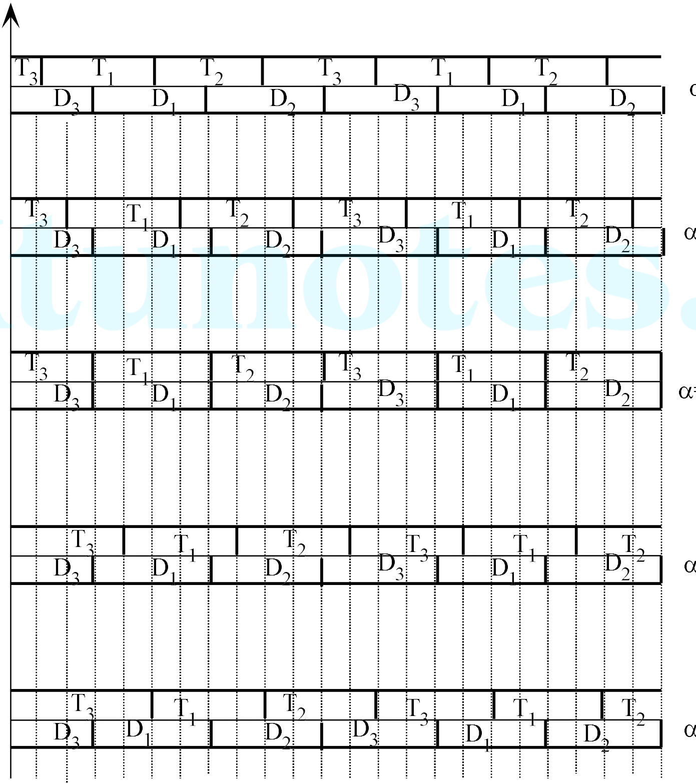


The circuit has continuous operation (load voltage is continuous) for α<
operation for α>60°.

α = 90°

At α = 90° (ωt=120°) T_1 is turned on. At that instant D_1 is forward biased,
through T_1 and D_1, and load voltage is equal to v_{AC}. At ωt=210°, D_2 becomes
is transferred from D_1 to D_2. Now, T_1 and D_2 conduct simultaneously to provid
load current. As a result, output voltage drops to zero. It continues until ωt=
on. Now load current flows through T_2 and D_2, and output voltage is made equa
Note : The load voltage has only 3 pulses per cycle.

α = 90°		
0-90°	T_3D_3	v_{CB}
90°-120°	T_3D_1	0
120°-210°	T_1D_1	v_{AC}
210°-240°	T_1D_2	0
240°-330°	T_2D_2	v_{BA}
330°-360°	T_2D_3	0



Note : If α>60°, output voltage is discontinuous.

AVERAGE OUTPUT VOLTAGE

For $\alpha \leq 60°$, during one pulse, T_1D_3 and T_1D_1 conducts, both $v_o=v_{AB}$ and $v_o= v_A$

For $\alpha > 60°$, for one pulse, T_1D_1 and T_1D_2 conducts, $v_o= v_{AC}$ only occurs.

Note : If $v_A = V_m \sin(\omega t)$ & $v_B = V_m \sin(\omega t - 120°)$, $v_{AB} =$

$v_{AC} = \sqrt{3} V_m \sin(\omega t - 30°)$

For $\alpha \leq 60°$,

Average value of output voltage is

$$\frac{\quad}{2\pi}(1+\cos\alpha)$$

3-phase semiconverter can operate only in one quadrant since average load vo
for all values of α.